- *Compliance*—You might face compliance challenges with data services more than with cloud native applications. Persisted data is usually critical for businesses and often contains information protected by specific laws, regulations, or customer agreements regarding how it's managed. For example, when dealing with personal and sensitive information, it's vital that you manage data in accordance with privacy laws. In Europe, that would mean following the General Data Protection Regulation (GDPR). In California, there is the California Consumer Privacy Act (CCPA). In other domains, further laws apply. For example, health data in the United States should be handled in compliance with the Health Insurance Portability and Accountability Act (HIPAA). Both the cloud native storage and cloud provider should comply with whatever laws or agreements you are required to respect. Because of this challenge, some organizations dealing with very sensitive data, like health care providers and banks, prefer to use a type of cloud native storage on their premises so they have more control over data management and can ensure compliance with the applicable regulations.

CATEGORIES OF DATA SERVICES

Data services can be categorized based on who is responsible for them: the cloud provider or you. Cloud providers have multiple offerings for data services, addressing all the main challenges of cloud native storage.

You can find industry-standard services like PostgreSQL, Redis, and MariaDB. Some cloud providers even offer enhancements on top of them, optimized for scalability, availability, performance, and security. For